



MINISTÈRE
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institut
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APPLICATION FORM IUF 2023-2028

INNOVATION CHAIR

Project that aims to transfer and promote research work to socio-economic world



Avant de remplir le formulaire de candidature,
nous vous conseillons de lire attentivement le document :
« **Notice de procédure de dépôt en ligne des candidatures** »



The application will be considered only if :

- Correctly and fully completed in French and English,
- Submitted by the deadline,
- Preferably saved in pdf format and named :
**formulaire_candidature_Innovation.pdf (version française) et
application_form_Innovation.pdf (version anglaise)**
- Using text font Arial (11)

APPLICATION TYPE

☐ JUNIOR

☒ SENIOR

NUMBERS OF PREVIOUS APPLICATION(S) / YEAR(S): 0

PREVIOUS IUF NOMINATION(S) / PROMOTION(S): 0

1) APPLICANT'S IDENTITY

☒ SIR

☐ MADAM

LAST NAME: Caumon

FIRST NAME: Guillaume

STREET ADDRESS: 84 rue du Général Custine

POSTCODE CITY: 54000

CITY: Nancy, France

PHONE: +33 6 79 81 85 75

E-MAIL ADDRESS: Guillaume.Caumon@univ-lorraine.fr

AGE ON 2023.01.01: 46

☒ PROFESSOR

☐ ASSOCIATE PROFESSOR

☐ PROFESSOR-HOSPITAL DOCTOR

☐ ASSOCIATE PROFESSOR-HOSPITAL DOCTOR

UNIVERSITY AFFILIATION: Université de Lorraine

SCIENTIFIC FIELD: Earth sciences, universe, space

FIELD ACCORDING TO CNU: Physics and Chemistry of the Earth (35)
SECTION CNU 2 :

IF APPLICABLE,

RESEARCH GROUP: GeoRessources

2) RESEARCH AND INNOVATION PROJECT

TITLE OF THE RESEARCH PROJECT:

SaMoA : **S**ampling geological **M**odels **A**daptively for geophysical inverse problems.

SUMMARY OF THE RESEARCH AND INNOVATION PROJECT (MAXIMUM 500 CHARACTERS) :

The goal is to propose new geological samplers which can honour spatial data and prior geological concepts while controlling the distance between the generated realizations. Controlling distances between models during sampling will allow for adaptively exploring the geological parameter space, including the number and connectivity of faults or layers. This will positively impact the effectiveness of geophysical inverse methods.

3) SCIENTIFIC ACTIVITY AND RESEARCH VALORISATION ACTIVITY

CURRENT SCIENTIFIC ACTIVITY (MAXIMUM 500 CHARACTERS):

I am doing research on multidisciplinary knowledge and subsoil data integration for the modeling and quantification of subsurface uncertainties. I am currently working on:

- The correlation of stratigraphic sections using interpretive concepts
- The characterization and reduction of structural uncertainties
- The updating and simplification of conforming meshes representing geological objects
- The management of discrete parameters in inverse problems.

PARTNERSHIP RESEARCH ACTIVITY, MATURATION OR TECHNOLOGY TRANSFER ACTIVITY (MAXIMUM 1500 CHARACTERS):

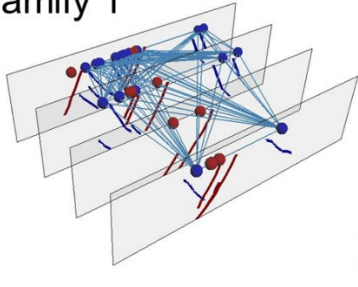
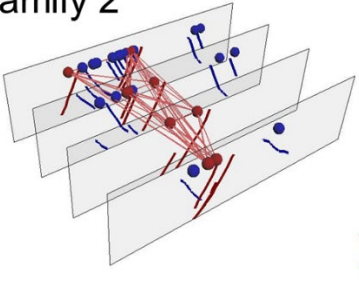
I am the director of the RING (Research for Integrative Numerical Geology) Consortium, which performs methodological research for subsurface modelling and its applications for the safe and reliable management of natural resources and assessment of natural hazards. The current industrial sponsors are ANDRA, AspenTech, Caisse Centrale de Réassurance (CCR), Chevron, Equinor, IFPEnergiesNouvelles, Orano, Repsol, Schlumberger, TotalEnergies, and 28 geological surveys (<https://www.ring-team.org/consortium>). All sponsors have access to the computer codes developed in the frame of the Consortium. These partners contributed to ca. 300 K€ of the consortium budget in 2021, including their consortium fee plus specific projects for PhD students working on statistical earthquake hazard modelling from CCR, uncertainty management in well correlation from Equinor, and local remeshing from TotalEnergies. The startup Geode-Solutions was founded in 2019 by two of my former PhD students from the open source project RINGMesh (now renamed OpenGeode). I was part of the scientific board for the Advanced Mining Technology Center in Chile in 2012-2015, and am currently an expert for the Scientific Council of IFPEnergiesNouvelles and for the BRGM GEOMOD program.

Examples of items to be mentioned: participation in a maturation or technology transfer project (transfer through licensing or company creation, for example in connection with a technology transfer organisation or incubator), participation in a Proof of concept project funded by the ERC or a SATT, participation in a Horizon Europe collaborative project, participation in a collaborative research project with a company in France. All scientific sectors are eligible. Also mention scientific competition activities or participation in the governance bodies of a company (e.g. board of directors, strategic steering committee,).

4) INDICATORS OF SCIENTIFIC PRODUCTION AND VALORISATION OF RESEARCH

PUBLICATIONS			
	ENTIRE CAREER	LAST 10 YEARS	LAST 5 YEARS
NUMBER OF ARTICLES IN PEER-REVIEWED JOURNALS	67	53	24
NUMBER OF BOOKS	0	0	0
NUMBER OF CHAPTERS IN EDITED BOOKS	5	3	1
NUMBER OF INVITED LECTURES	14	7	5
NUMBER OF PATENTS*, SOFTWARE AND KNOW-HOW REGISTERED	0	0	0
NUMBER OF PATENTS*, SOFTWARE AND KNOW-HOW EXPLOITED	3	3	2
NUMBER OF RESEARCH COLLABORATION CONTRACTS WITH COMPANIES AND LOCAL AUTHORITIES	32	24	17
NUMBER OF CONSULTANCY CONTRACTS WITH COMPANIES OR LOCAL AUTHORITIES	3	3	2

DETAILS OF THE 3 MOST SIGNIFICANT PUBLICATIONS AND 3 PATENTS*, SOFTWARE AND/OR KNOW-HOW (4 PAGES MAXIMUM)	
TITLE	1. Formal validity criteria of a structural geological model
REFERENCE	Caumon, G. , Lepage, F., Sword, C. H., and Mallet, J.-L. (2004). Building and editing Sealed Geological Models. <i>Math Geol</i> , 36 (4):405-424.
SUMMARY	In this paper, we propose some necessary conditions for a three-dimensional surface-based sealed geological model to be valid. Namely, such a model should have all the characteristics of a valid boundary representation, plus specific constraints depending on the geological type of interfaces: layer boundaries cannot cross, and only fault surfaces may have free boundaries not terminating on other surfaces. From these validity criteria, we proposed model building and editing algorithms, which exploit the notion of surface parameterization to simplify the management of branch lines between geological surfaces.
TITLE	2. Managing uncertainties on the number and connectivity of fault surfaces for the resolution of a hydrodynamic inverse problem.
REFERENCE	
SUMMARY	This paper investigates about the ability of hydrodynamic data to discriminate between various possible fault network scenarios anchored on sparse geological sections. For this, we use a parameterization for stochastic fault networks proposed by Cherpeau et al (2010) where each fault is sequentially generated by a few monte Carlo samplings of the fault position and geometry, and then represented implicitly as a level set of a signed distance function. In the resulting fault networks, the fault branching relationships emerge from the simulation order as younger faults terminate on older ones. This process is combined with a conditioning approach so that the simulated faults honor sparse observations. We use a gradual deformation approach to explore the flow response in the neighborhood of several structural scenarios generated by the fault sampling method: for each scenario, small perturbations to the random numbers used in the fault simulation allow to perturb the fault network. This process is applied in a Monte Carlo Markov Chain process to gradually change each scenario until the hydrodynamic mismatch reaches an acceptable threshold.
TITLE	3. Review and conceptualization of the relations between, geometrical and petrophysical parameters in a geological model
REFERENCE	Caumon, G. (2018). Geological Objects and Physical Parameter Fields in the Subsurface: A Review. In: Daya Sagar BS, Cheng Q, Agterberg F (eds) Handbook of Math. Geosci., Springer, pp 567–588
SUMMARY	In this paper, I review the relationships between geological interpretations and the descriptions used by geophysicists to describe the subsurface. This initial formulation is further developed in the companion paper written with Florian Wellmann in <i>Advances in Geophysics</i> in 2018. TODO

TITLE	4. FaultMod Software for hierarchical generation of fault scenarios from sparse data.
REFERENCE	Godefroy, G., Caumon, G., Laurent, G., Bonneau, F. (2021). Multi-scenario interpretations from sparse fault evidence using graph theory and geological rules. JGR: Solid Earth. 126(2):e2020JB020022.
SUMMARY	In this work realized with my PhD student Gabriel Godefroy, we propose an innovative algorithm to generate fault scenarios based on sparse subsurface interpretations inferred from seismic sections or borehole data. Family 1  Family 2  <i>Figure 1 : Possibility graphs for associating two families of fault traces in 2D sections. The computation of maximal cliques of this graph is used to design a stochastic association sampler.</i> The two main novelties concern: - The use of graphs and geological rules to represent the likelihood that two distant observations belong to the same fault object. This representation makes it possible to integrate various types of observations and geological knowledge. Among others, the notion of graph cliques opens the way to defining complex interactions multiple fault observations (See Task 1.1). - The introduction of a hierarchical sampling algorithm which tends to process important faults
TITLE	5. WeCo Software for hierarchical generation of fault scenarios from sparse data.
REFERENCE	Caumon, G., Antoine, C. (2019). Stochastic stratigraphic correlation of multiple wells seen as a Directed Acyclic Graph creation problem. IAMG 2019, Penn State.
SUMMARY	
TITLE	6.
REFERENCE	
SUMMARY	

***In case of patents**, specify whether you are the holder or co-owner and in which field. **Give the title, number and filing date of the patent.**

5) PHD SUPERVISION

PHD SUPERVISION	ENTIRE CAREER	LAST 5 YEARS
NUMBER OF SUPERVISED THESES (COMPLETED) Specify the number of theses co-financed with a company or a public authority, including CIFRE theses*	26 - 4 CIFRE (2 Total, 1 Schlumberger, 1 CGG) - 19 RING Consortium - 1 RING + Equinor - 2 RING + Chevron	10 - 2 CIFRE (Total, Schlumberger) - 7 RING Consortium - 1 RING + Equinor
NUMBER OF SUPERVISED ONGOING THESES Specify the number of theses co-financed with a company or a public authority, including CIFRE theses *	Currently: 7 - 1 research contract (CCR) - 4 RING Consortium - 1 RING + TotalEnergies - 1 Ministry / Region Grand Est.	

*Industrial convention for training through research (via the ANRT: Agence nationale de la recherche et de la technologie). **For each CIFRE thesis, specify the partner.**

Detailed list of PhD supervisions (last 5 years):

Name (date)	Topic	Co-advisor(s)	Funding	Job after the PhD
Giusi Ruggiero (09/2022-09/2025)	Stochastic inversion of FWI images for reducing structural uncertainty	Paul Cupillard (UL)	RING Consort.	Ongoing
Paul Marchal (06/2022-06/2025)	Structure-guided modelling of hydrothermal alteration. Application to unconformity Uranium deposits	Pauline Collon (UL) Julien Mercadier (CNRS)	Orano (GeoMin3D)	Ongoing
Julien Herrero (01/2022-01/2025)	Transdimensional inversion of porous flow data	Thomas Bodin (ENS Lyon) Mustapha Zakari (CNRS)	RING Consort.	Ongoing
Marius Rapenne (12/2021-12/2024)	Adaptive homogeneization and meshing for seismic hazard assessment.	Paul Cupillard (UL) Corentin Gouache (CCR)	CCR	Ongoing
Fabrice Taty-Moukati (03/2021-03/3024)	Stochastic interpretation of faults on seismic images with marked point processes.	Radu Stoica (UL)	RING Consort.	Ongoing
Enrico Scarpa (01/2021-01/2024)	Bridging the gap between static and flow models for complex channelized reservoirs	Pauline Collon (UL)	RING Consort.	Ongoing
Zoé Renat (12/1019-12/2022)	Time reversal for microseismic characterization	Paul Cupillard (UL)	MSER Region Grand Est	Ongoing

Capucine Legentil (12/1019-12/2022)	Local tetrahedral model updating: geometry and topology	Paul Cupillard (UL) Jeanne Pellerin (TotalEnerg.)	RING Consort. TotalEnerg.	Ongoing
Paul Baville (01/2019-04/2022)	Stratigraphic correlation uncertainty: impact of transport directions on computer-based multi-well correlations	Cédric Carpentier (UL) Marcus Apel (Equinor)	RING Consort. Equinor	Postdoc search (Germany)
Melchior Schuh-Senlis (02/2018-03/2021)	Using Stokes flow equations for the restoration of geological structural models	Paul Cupillard (UL)	RING Consort.	Geode-Solutions (research engineer)
Nicolas Mastio (02/2017-04/2020)	Improving the global coherency of the Shared Earth Model using static, dynamic and geomechanics data	Pierre Thore (TotalEnerg.) Marianne Conin (UL)	CIFRE TotalEnerg.	CapGemini (software developer)
Nicolas Clausolles (02/2017-03/2020)	Stochastic seismic interpretation of salt bodies: detection, sampling and impact on seismic imaging	Pauline Collon (UL)	RING Consort.	BRGM (geoscientist)
Pierre Anquez (02/2016-04/2019)	Repair and simplification of 3D structural models: impact on meshing and numerical simulations in seismology and hydrodynamics	Bruno Lévy (inria) Jeanne Pellerin (Total Energ.)	RING Consort.	Geode-Solutions (CEO)
Modeste Irakarama (12/2015-04/2019)	Structural uncertainty reduction by waveform inversion	Paul Cupillard (UL) Paul Sava (CSM)	RING Consort.	Exxonmobil Upstream Research (scientist)
Julien Renaudeau (03/2016-04/2019)	Meshless structural modeling and restoration	Bruno Levy (inria) Frantz Maerten (SLB)	CIFRE Schlumberger	SLB (Project Manager)
Marion Parquer (10/2014-04/2018)	Data and trend conditioning in pseudo-genetic simulation of sedimentary bodies	Pauline Collon (UL)	RING Consort.	NR Canada (Postdoc)
Gabriel Godefroy (11/2014-03/2018)	Uncertainty and kinematic concepts in structural modeling	Mary Ford (UL)	RING Consort.	Total GRC (Research scientist)

6) CURRICULUM VITAE

Guillaume Caumon

Professor (PR1)

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Guillaume.Caumon@univ-lorraine.fr

Born 12/28/1976 (45 years old)

2 daughters (13 and 17)

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<http://ring.georessources.univ-lorraine.fr>

Research interests

Development of methodologies to integrate geological and physical observations in a common theoretical and numerical framework.

Positions held

- 02/2011-today **Professor (PR1) in Numerical Geology at Université de Lorraine**, Teaching at *École Nationale Supérieure de Géologie (ENSG)* and *GeoRessources (UMR7359)*
- 01/2007-today **Director of RING Consortium (formerly Gocad Research Consortium).**
- 04-07/2015 **Visiting Professor**, Center for Wave Phenomena, Colorado School of Mines
- 09/2004-01/2011 **Maître de Conférences in Numerical Geology (Associate Professor) at INPL (now Université de Lorraine)**, ENSG and CRPG (UPR 2300 CNRS).
- 08/2003-08/2004 **Postdoctoral Scholar**, *Earth Resources Eng. Dept.*, Stanford Univ. Sponsor: Chevron ETC.
- 09/1999-07/2003 **Research and teaching assistant**, INPL, CRPG. Scholarship: Chevron ETC
- 02/1999-12/2002 **Research intern, Chevron ETC, (total duration: 11 months)**

Education

- 10/2009 **Habilitation diploma (HDR), INPL.**
- 03/2003 **PhD, Geosciences.** Sponsored by Chevron ETC
- 1999 **MSc: DEA (INPL) and Diplôme d'Ingénieur de l'École Nationale Supérieure de Géologie.**

Research outputs

67 journal publications

5 book chapters

58 peer-reviewed conference papers

14 invited and keynote conferences

Orcid: [0000-0002-7828-4600](https://orcid.org/0000-0002-7828-4600)

Bibliometrics (10/2022)	Citations	Citations (w/o self-citations)	H index
WoS	1545	1239	22
Google Scholar	3454	N/A	31

Research supervision

- Supervision of over 45 MSc students since 2004.
- Supervised 26 PhD students (see details below)
- External PhD committee member for 30 PhD students since 2010

Main research responsibilities

- 2007-to-date: Director of the RING (Research for Integrative Numerical Geology) Consortium. 10 industrial sponsors, 28 geological surveys, 84 universities. Annual budget (2021): 380 K€.
- 2012-to-date: in charge of the Integrative Numerical Geology team of GeoRessources and member of the directory committee.
- 2008-to-date: Editorial board member of *Mathematical Geosciences*

- 09/2022: Conference chair of *IAMG 2022*, the Annual Meeting of the International Association of Mathematical Geosciences. Nancy, 300 participants.
- 02/2021: Organization committee of the 2nd AAPG Middle East Workshop on decision-based Integrated Reservoir Modeling.
- 2016, 2017: Special section guest editor for the SEG/AAPG journal *Interpretation*.
- Reviewer for Mathematical Geosciences, Computers & Geosciences, Geoscientific Model development, Solid Earth, Basin Research.

Committees, support to innovation

- 08/2022-to-date: Member, Scientific Committee of the GEOMOD Program (BRGM)
- 04/2020-to-date: Expert for the Scientific Council of IFPEN
- 2017-to-date: Member, experts committee of *Pole Avenir* (French subsurface competitiveness cluster)
- 06/2019: Member of the assessment board of the Geosciences Dept. of IFPEN
- 05/2018: Member of the assessment panel of the Niels Bohr Institute, Copenhagen, Denmark
- 2012-2014: Scientific Advisory board, Advanced Mining Technology Center, Chile (www.amtc.cl)

The RING Consortium, which I direct, supports all its industrial members with new technologies (see Sections 3, 7). RING is also a partner of three startup companies:

- Geode-Solutions, created by two of my former PhD students: <https://geode-solutions.com/>
- Tessaël: <https://www.tessaël.com/>
- Geotref Venture: <https://www.polytechnique.edu/innovation/start/geotref-venture>

Distinctions

- *Chevalier des Palmes Académiques* from the French Ministry of Education (2020).
- Michel Gouilloud Schlumberger Award from the French Academy of Sciences (2019).
- SPE Regional Distinguished Achievement Award for Petroleum Engineering Faculty for South, Central and East Europe Region (2016).
- *Computers & Geosciences* 2015 Best Paper Award (Laurent, Caumon & Jessell)
- *Computers & Geosciences* 2014 Best Paper Award (Pellerin, Lévy & Caumon)
- Vistelius Award of the International Association of Mathematical Geosciences (2009).
- Significant Achievement Award from Society of Exploration Geophysicists (SEG) to the Gocad Research Consortium (2009).
- Texas Instrument Student paper award to PhD student Luc Buatois, High Performance Computing Conference (2007).

Teaching

Approx. 220 contact hours annually at Université de Lorraine:

- Geostatistics: Stationarity and geology, spatial variability, estimation, simulation (M1-S7, since 2012)
- C++ programming (M2-S9, since 2004)
- Static reservoir modeling: from seismic interpretation to flow simulation. (M1-S8, since 2008)
- 3D modeling of ore deposits: Mineral targeting and resource assessment (M2-S9, 2006-2016)
- Subsurface uncertainties (M2-S9, since 2004)
- Geomathematics: differential geometry, meshes, implicit structural modeling (M2-S9, since 2007)
- Petroleum Field Case (in partnership with TotalEnergies, M2-S9, since 2005)
- Introduction to Data management in Geosciences (M1 S8; Telecom Nancy, since 2016)
- Field mapping and Geomodeling: 3D structural modeling from student's field data (L3-S6, 2008-2020)
- Petroleum Geoscience field trip: connecting field observations with reservoir modeling strategies (M1-S8, 2011-2019)

Teaching responsibilities

- Responsible of the Numerical Geology specialty (S9) since 2006: definition of the academic program in conjunction with industry needs, choice of instructors, personalized mentoring of students. Member of the ENSG Direction committee.
- Elected Member of the ENSG council (2006-2010 and 2014-2020).
- Mentoring undergraduate and graduate students (SPE chapter faculty advisor since 2010, AAPG IBA tutor 2012-2016, projects, teaching assistants)

7) PERSONAL BIBLIOGRAPHY

Names *in italic* correspond to advised graduate students (See Section 5 for details).

* indicates papers with accompanying open source code. The **code name** is given after the reference

1. Balarac, G., Basile, F., Bénard, P., Bordeu, F., Chapelier, J.-B., Cirrottola, L., **Caumon, G.**, Dapogny, C., Frey, P., Froehly, A., Ghigliotti, G., Laraufie, R., Lartigue, G., *Legentil, C.*, Mercier, R., Moureau, V., Nardoni, C., Pertant, S., Zakari, M. (2022). Tetrahedral remeshing in the context of large-scale numerical simulation and high performance computing. *MathematicS In Action*. 11:129-164.
2. Irakarama, M., Thierry-Coudon, M., Zakari, M., **Caumon, G.** (2022). Finite element implicit 3D subsurface structural modeling. *Computer-Aided Design*. 149:103267.
3. **Legentil, C.*, Pellerin, J., Cupillard, P., Froehly, A., **Caumon, G.** (2022). Testing scenarios on geological models: Local interface insertion in a 2D mesh and its impact on seismic wave simulation. *Computers & Geosciences* 159:105013. **Lumos**
4. **Anquez, P.*, Glinsky, N., Cupillard, P., **Caumon, G.** (2022). Impacts of geometric model simplifications on wave propagation—application to ground motion simulation in the lower Var valley basin (France). *Geophysical Journal International* 229, 110–137. **Scar**
5. *Baville, P.*, Apel, M., Hoth, S., Knaust, D., Antoine, C., Carpentier, C., **Caumon, G.** (2022). Computer-assisted stochastic multi-well correlation: Sedimentary facies versus well distality. *Marine and Petroleum Geology* 135:105371.
6. *Grose, L., Ailleres, L., Laurent, G., **Caumon, G.**, Jessell, M., Armit, R. (2021). Modelling of faults in LoopStructural 1.0*. *Geosci. Model Dev.* 14, 6197–6213.
7. Godefroy, G., **Caumon, G.**, Laurent, G., Bonneau, F. (2021). Multi-scenario interpretations from sparse fault evidence using graph theory and geological rules. *Journal of Geophysical Research: Solid Earth*. 126(2):e2020JB020022.
8. *Irakarama, M.*, Laurent, G., Renaudeau, J., **Caumon, G.** (2021). Finite Difference Implicit Structural Modeling of Geological Structures. *Mathematical Geosciences*.
9. Martinelli, M., Bistacchi, A., Mittempergher, S., Bonneau, F., Balsamo, F., **Caumon, G.**, Meda, M. (2020). Damage zone characterization combining scan-line and scan-area analysis on a km-scale Digital Outcrop Model: The Qala Fault (Gozo). *Journal of Structural Geology*, 140, 104144.
10. *Schuh-Senlis, M.*, Thieulot, C., Cupillard, P., **Caumon, G.** (2020). Towards the application of Stokes flow equations to structural restoration simulations. *Solid Earth*, 11, 1909–1930.
11. *Mastio N.*, Thore, P., Conin, M., **Caumon, G.** (2020). Determination of a stress-dependent rock-physics model using anisotropic time-lapse tomographic inversion. *Geophysics*, 85(4), C141–C152.
12. Murlanette, P., Biver, P., Renard, P., Noëtinger, B., **Caumon, G.**, Perrier, Y. A. (2020). Direct simulation of non-additive properties on unstructured grids. *Advances in Water Resources*, 143, 103665.
13. *Parquer, M.*, **Caumon, G.**, Collon, P. (2019). A stochastic reconstitution method for determining the abandoned channel meander chronology. *Geomorphology*.
14. *Clausolles, N.*, Collon, P., & **Caumon, G.** (2019). Generating variable shapes of salt geobodies from seismic images and prior geological knowledge. *Interpretation*, 1–45.
15. *Renaudeau, J.*, Malvezin, E., Maerten, F., **Caumon, G.** (2019). Implicit structural modeling by minimization of the bending energy with moving least squares functions. *Mathematical Geosciences*. 51(6):693-724.
16. Godefroy, G., **Caumon, G.**, Laurent, G., Bonneau, F. (2018). Structural interpretation of sparse fault data using graph theory and geological rules. *Mathematical Geosciences*.
17. *Anquez P.*, Pellerin J., *Irakarama M.*, Cupillard P., Lévy B., **Caumon G.** (2019). Automatic correction and simplification of geological maps and cross-sections for numerical simulations. *Comptes-Rendus Geosciences*. 351:48-58.
18. *Bardy, G.*, Biver, P., **Caumon, G.**, Renard, P. (2019). Oil production uncertainty assessment by predicting reservoir production curves and confidence intervals from arbitrary proxy responses. *Journal of Petroleum science and Engineering* 176:116.
19. *Irakarama M.*, Cupillard P., **Caumon G.**, Sava P. & *Edwards J.* (2019). Appraising structural interpretations using seismic data - theoretical elements. *Geophysics*.
20. Wellmann, F. & **Caumon, G.** (2018). 3-D Structural Geological Models: Concepts, Methods, and Uncertainties. *Advances in Geophysics*. 59: 1-121.

21. Edwards, J., Lallier, F., **Caumon, G.**, Carpentier, C. (2018). Uncertainty management in stratigraphic well correlation: A training based method. *Computers & Geosciences*. 111:1-17.
22. Grose L., Laurent G., Aillères L., Armit, R., Jessell, M., **Caumon, G.** (2017). Structural data constraints for implicit modeling of folds. *Journal of Structural Geology* 104:80–92.
23. Godefroy, G., **Caumon, G.**, Ford, M. (2018). A parametric fault displacement model to introduce kinematic control into modeling faults from sparse data. *Interpretation*. 6(2), B1-B13.
24. Chauvin, B. P., Stockmeyer, J. M., Lovely, P. J., Plesch, A., **Caumon, G.**, Shaw, J. H. (2018). Consistent boundary conditions in 3D mechanics-based restoration: validation on an extensional sandbox model. *AAPG Bulletin* 102(2):245-266.
25. Parquer, M. N., Collon, P., **Caumon, G.** (2017). Reconstruction of channelized systems through a conditioned reverse migration method. *Mathematical Geosciences* 49(8):965-994.
26. *Pellerin, J., Botella, A., Mazuyer, A., Chauvin, B., Bonneau, F., Levy, B., **Caumon, G.** (2017). RINGMesh: A programming library for developing mesh based geomodeling applications. *Computers & Geosciences*. 104:93-100. **This paper corresponds to the RINGMesh open-source code, which has led to the creation of the Geode-solution startup.**
27. Wu, X., **Caumon, G.**, (2017). Simultaneous multiple well-seismic ties using flattened synthetic and real seismograms. *Geophysics* 82(1):IM13-IM20.
28. Bonneau, F., **Caumon, G.**, Renard, P. (2016). Impact of a Stochastic Sequential Implantation of Fractures on the Spatial Correlations and Connectivity of Discrete Fracture Networks. *Journal of Geophysical Research – Solid Earth*. 121(8): 5641–5658
29. Botella, A., Levy, B., **Caumon, G.** (2016). Indirect unstructured hex-dominant mesh generation using tetrahedra recombination. *Computational Geosciences* 20(3):437-451.
30. Collon, P., Pichat, A., Kergaravat, C., Botella, A., **Caumon, G.**, Ringenbach, J.-C., Callot, J.-P. (2016). 3D modeling from outcrop data in a salt tectonic context: Example from the Inceyol minibasin, Sivas Basin, Turkey. *Interpretation* 4:SM17-SM31.
31. Lallier, F., **Caumon, G.**, Borgomano J., Viseur, S., Royer, J.-J. (2016). Uncertainty assessment in stratigraphic well correlation of a carbonate ramp: method and application to the Beausset Basin, SE France. *CR Géosciences* 348(7):499-509.
32. Laurent, G., Aillères, L., Grose, L., **Caumon, G.**, Jessell, M., Armit, R. (2016). Implicit modeling of folds and overprinting deformation. *Earth and Planetary Science Letters* 456:26–38.
33. Ramón, M. J., Pueyo, E. L., **Caumon, G.**, Briz, J. L. (2016). Parametric unfolding of flexural folds using palaeomagnetic vectors: Geological Society, London, Special Publications, 425:247-258.
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1. **Caumon, G.**, Collon, P. (2022). Modélisations structurales implicites ou explicites et leurs rôles pour la modélisation rapide de formations géologiques. *Transitions numériques en géologie de l'ingénieur*, CFGI, Paris. **Invited presentation**
2. **Caumon, G.** (2022). Association de données éparses (failles et/ou stratigraphie) : nouvelles approches de corrélation et de modélisation des incertitudes. *Transitions numériques en géologie de l'ingénieur*, CFGI, Paris. **Invited presentation**
3. Giraud, J., **Caumon, G.**, Grose, L., Cupillard, P. (2022) Geometrical Inversion Coupled with Automated Geological Modelling. *83rd EAGE Annual Conference & Exhibition*. p. 1-5. European Association of Geoscientists & Engineers, Madrid, Spain / Online, Spain. [DOI:10.3997/2214-4609.202210522](https://doi.org/10.3997/2214-4609.202210522)
4. **Caumon, G.** (2021). Behind the outcrop: On 3D subsurface modeling and uncertainty management. *VCG 2021*, Marseille, France and online. **Keynote presentation**
5. **Caumon G.** (2021) A (biased) historical review and status of 3D geological modeling, IMAGE workshop 09 : 3D Computer Geologic Modeling for Geophysicists and How to Integrate Geologic and Geophysical Computer Earth Modeling. **Invited presentation**
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12. **Caumon, G.**, Godefroy, G., Edwards, J. (2018). On the Need for Spatial adaptivity and geological constraints in integrative subsurface models. EAGE Workshop 02 (Data Integration in Geoscience - Perspectives for Computational Methods), Copenhagen, Denmark. **Keynote presentation**
13. Clausolles, N., Collon, P., **Caumon, G.** (2018). A Workflow for 3D Stochastic Modeling of Salt from Seismic Images. In 80th EAGE Conference and Exhibition 2018. Copenhagen: EAGE. DOI:10.3997/2214-4609.201801272
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Software uses, derivative works and industrialization:

I have been the director of the RING (formerly Gocad) Consortium since 2007. The goal of this consortium is to develop new approaches and software for integrative subsurface modeling. In exchange for their financial contribution, the academic and industrial members of the RING Consortium gain an access to software developed by the RING team, with the right to reuse and modify. Industrial members also have the right to exploit RING software and create proprietary derivative work, but they do not necessarily communicate on this. Below is a list of software transfers and some examples of use by socio-economic stakeholders. More references are available at <https://www.ring-team.org/technologies> and <https://www.ring-team.org/home/about/26-industrialization-of-our-prototypes>.

In addition, all the software written by the CIFRE PhD students I have advised have also been integrated in proprietary codes of the sponsoring companies (CGG, TotalEnergies, Schlumberger).

1. **StructuralLab** (<https://www.ring-team.org/software/skua-gocad-plugins/42-structurallab>) is a plugin of the Gocad software for implicit structural modeling. An open-source version has been recently reimplemented in the LoopStructural open-source package (Grose et al., *Geosci. Model Devel.* 2021). LoopStructural also includes a kinematic model of near field fault displacement (Grose et al., 2021), which was originally developed with my PhD students Gautier Laurent (Laurent et al., *Tectonophys.* 2014) and Gabriel Godefroy (Godefroy et al., *Interp.* 2018).
2. **RINGMesh** (<http://ringmesh.org/>) is a stand-alone open-source library to promote interoperability and provide useful tools for using unstructured meshes in geoscience applications (Pellerin et al, *Comp. & Geosci.* 2017). It was initiated by a team of RING PhD students in 2014. In 2019, the startup Geode-solution (<https://geode-solutions.com/>) was created by two of my former PhD students (Arnaud Botella and Pierre Anquez). The company took over the project, its maintenance and evolution. They rebranded the library as OpenGeode.
3. **UncertaintyLab** was a plugin of the Gocad software which I developed during my postdoc at Stanford. It included an implementation of geostatistical simulation on unstructured meshes (Remy, Caumon & Levy, *Gocad Meeting*, 2004). In 2006 and 2007, we applied it to sample structural uncertainties on implicit stratigraphic horizons (Caumon & Mallet, *IAMG 2006*; Caumon, Tertois and Zhang, *EAGE Petrol. Geostat.*, 2007). This contributed to the SKUA Structure and uncertainty workflow, which is part of the SKUA software suite (Mallet and Tertois, 2010), and was reimplemented in the FaultMod plugin for stochastic structural modeling (Cherpeau et al, *Math. Geosci.* 2012; Cherpeau & Caumon, *Petrol. Geosci.* 2015).
4. **ODSIM** was a plugin of the Gocad software developed with my PhD student Vincent Henrion in 2007-2009 to model karsts (Henrion et al, *Geostat. Congress 2008*; Henrion et al, *Math. Geosci.* 2010). It was taken over by my colleague Pauline Collon and her students in the 2010's to push the idea further

and create in a derivative code named KarstMod ([Collon-Drouaillet et al., GSL Spec. Pub. 2012](#)). In 2019, this plugin was industrialized by Petrobras and Emerson ([Backheuser et al, RING Meeting 2019](#)).

5. **SCAR** is a mesh repair and simplification code developed by my PhD student Pierre Anquez. This code was recently used to reduce simulation time in seismic hazard assessment, in partnership with CEREMA (Anquez et al, Geophys. J. Int. 2022).
6. **FaultMod** is a plugin of the Gocad software for stochastic fault modeling, developed with my PhD students Nicolas Cherpeau, Charline Julio and Gabriel Godefroy. It has been used and extended by TotalEnergies to address structural uncertainties ([Lallier et al, EAGE Int. Res. Mod. 2014](#); [Lallier & Clausolles, EAGE 2017](#); Yong et al, EAGE; [Godefroy et al, EAGE 2019](#)). The most recent version includes the first implementation of a graph-based fault association sampling algorithm.
7. **WeCo** is a stand-alone multi-well correlation code based on Dynamic Time Warping, developed with Christophe Antoine (Research Engineer of the RING Consortium). This plugin extends prior work done with my PhD students F. Lallier and Jonathan Edwards in the Gocad plugin SCube to use of Dynamic time warping in stratigraphic correlation. Scube was used and extended by TotalEnergies ([Yoon et al, EAGE 2017](#)). WeCo is currently under consideration for industrialization by Equinor (Marcus Apel, maap@equinor.com, Pers. Com.).

8) IUF RESEARCH AND INNOVATION PROJECT 2023-2028

CHOICE OF AN IUF INNOVATION CHAIR

You are expected to submit a project of excellence in scientific research of the IUF standard, whatever the scientific field (in the natural and medical sciences or in the humanities and social sciences - Humanities in the broadest sense), which gives high priority to innovation in your field of research, in particular your own scientific results and discoveries linked to the IUF project.

Your project must aim to transfer and valorise research work to companies (existing or created on this occasion), local authorities, associations and citizens, and enable, through the application of research, to promote real innovation. The work concerned may be methodological and/or technological, disruptive or incremental. The projects identified must be likely to have measurable positive effects on the economy, society, culture, public policies or services of public interest.

Please specify the external partner(s) (precise contact details) of the IUF project for the period 2023-2028.

Motivation and state of the art

At a time where coal, petroleum and gas production must decrease to reduce CO₂ emissions and curb climate change, basins remain very important objects to study in applied geosciences. As billions of people live on or close to sedimentary basins, the characterization and protection of basinal groundwater resources is expected to take vital importance because of changing climate, increasing pollution and rising sea level; basins will also increasingly be used for geothermal energy, sequestration of CO₂ and confinement of nuclear waste; they host significant mineral resources; they can be locally very sensitive to seismic hazards because of quaternary sediments. Fundamentally, they also record essential elements to decipher the earth's history. However, this history is also a source of complexity of the rock material properties, which are heterogeneous and can display **abrupt changes along geological interfaces such as horizons and faults**. This makes subsurface forecasting a difficult task prone to uncertainty.

As direct observations are restricted to the surface of the earth and to highly localized drillholes, geophysics plays a key role to produce images of the subsurface. However, each geophysical method is sensitive to only specific physical properties (e.g., resistivity for magnetotelluric methods, elastic properties for seismic methods) and is affected by intrinsic physical limitations (e.g., decay with increasing observation distance, attenuation). Therefore, many questions about the Solid Earth, its structure, its evolution, and its natural resources rely on interpretation and modeling. Among these applications, examples include the monitoring and design of early warning systems for earthquakes or volcanic hazards, the design and operation of CO₂ sequestration in deep aquifers or depleted hydrocarbon reservoirs, the environmentally viable extraction of metals needed for the energy transition, the optimization of geothermal heat recovery, etc. To reach these goals, geophysical images (e.g., reflection seismic) are generally interpreted by geologists to produce a three-dimensional geological model, which in turn can be used for numerical forecasting of the physical behavior (e.g., coupled hydromechanical behavior of a CO₂ sequestration site).

However, several explanations for a given set of observations can generally be found in earth sciences (Chamberlin, 1931). This ambiguity has been much studied in geophysical inverse problems, which concern the identification of models whose physical response can explain a set of geophysical measurements (Tarantola, 2005). The non-unicity of the solution to inverse problems stems from the incompleteness and noise of these measurements, the lack of accuracy of the physical forecasting (forward problem) and the non-linearity of most geophysical processes (Hunt et al., 2007; Mosegaard and Sambridge, 2002; Snieder, 1998). Therefore, although many very interesting optimization approaches have been designed to solve geophysical inverse problems in a deterministic way, it is clear that a probabilistic

view of the problem is extremely useful to address the likely existence of multiple scenarios (Moorkamp et al., 2016; Mosegaard and Sambridge, 2002; Scheidt et al., 2018; Tarantola, 2006).

This probabilistic view has been much developed to find possible ranges of subsurface parameters for a particular subsurface model $\mathbf{m}$ from the data at hand $\mathbf{d}$ using a Bayesian formulation:

$$P(\mathbf{m}|\mathbf{d}) = \frac{P(\mathbf{d}|\mathbf{m})P(\mathbf{m})}{\int_{\mathbf{m}} P(\mathbf{d}|\mathbf{m})P(\mathbf{m})}.$$

In general, sampling the posterior probability $P(\mathbf{m}|\mathbf{d})$ is generally extremely challenging from a computational standpoint. Indeed, the likelihood term $P(\mathbf{d}|\mathbf{m})$ is derived from the difference between the observed data and the numerical forecast made from model $\mathbf{m}$ by numerical simulation, and must be normalized over all possible models. Several types of Markov chain Monte-Carlo methods have been proposed to approximate the posterior distribution by means of random walks in parameter space (Mosegaard and Sambridge, 2002; Tarantola, 2005), and numerous applications of these methods to geoscience problems have been proposed. The step size in the random walk (controlled by the proposal distribution) is difficult to choose, but is very important for the efficiency of the inverse problem resolution, as it determines how rapidly the algorithm converges. The notion of step size implies that a distance between samples is defined, which is also an essential concept in other inverse methods such the neighborhood algorithm (Sambridge, 2001), evolutionary algorithms (Pace et al., 2021) and Bayesian Evidential learning (Scheidt et al., 2018). Figure 1 illustrates the role of distances between models to approximate a response surface by a piecewise constant function centered on a limited set of sample models visualized in some abstract two-dimensional parameter space.

In addition to the mathematical techniques above, the importance of integrating geological concepts in defining the a priori distribution $P(\mathbf{m})$ has been much highlighted (Linde et al., 2015; Wood and Curtis, 2004). The model parameters $\mathbf{m}$ themselves are also defined from a combination of physical considerations (what do we expect to learn from the physical data at hand) and prior geological information (what could the medium of interest look like?). The latter question may concern the prior distribution of individual model parameters and the statistical description of the interactions between these parameters, which is very challenging when considering geological structures. It is also a fundamental geological question which relates to the translation of geological knowledge into a mathematical way. Although this task appears daunting because of the specific scientific approaches characteristics of geology (Frodeman, 1995), it is of very practical importance and is receiving more and more attention (Arnold et al., 2019; Caumon, 2018; Linde et al., 2015).

The question of how geological considerations should be integrated in geophysical inversion is often subject to debate. Indeed, geological interpretation can be biased and may be a source of subjectivity in the results. Also, the principle of parsimony calls for using the simplest possible model to explain the observations whereas geological parameterization may be seen as too complex. This justifies the use of “minimal priors” to generate images as objective as possible, which can then be interpreted as geologists. This approach, however, does not remove geological subjectivity (Bond et al., 2007), and it does not guarantee the compatibility between the final geological model and all physical observations. Downscaling of geophysical images constrained by a geological prior may appear as a possible way forward to make this modeling approach practical (Hedjazian et al., 2021). More fundamentally, considering forecasting and uncertainty quantification typically affects how to define what a “parsimonious model” means and advocates for integrating parameter-rich models in inverse problems (e.g., Hunt et al., 2007; Scheidt et al., 2018).

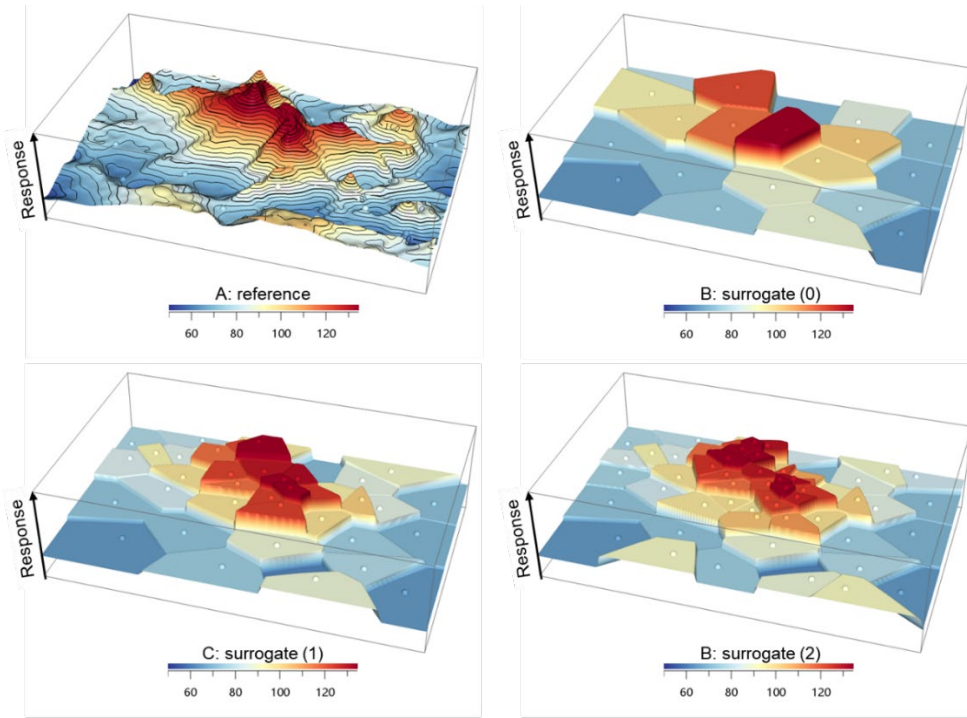


Figure 2 : Example of the piecewise constant approximation of a response surface representing for example the similarity between some observed geophysical data and numerical model forecasts in an abstract two-dimensional parameter space. A: Reference surface. B,C,D, successive approximations using the neighborhood algorithm (Sambridge, 2001) based on discrete sample models figured by the colored spheres.

Therefore, several authors have started to formalize geological concepts as complex priors for geophysical inverse problems (see recent reviews by Linde et al., 2015; Moorkamp et al., 2016; Wagner and Uhlemann, 2021; Wellmann and Caumon, 2018). Many of these approaches are based on complex random field models, such as those used in multiple point geostatistics (MPS) or those using local anisotropy information often termed structure-guided inversion. Both have been applied so far to spatial models having a homogeneous spatial resolution. To navigate in the parameter space, two very interesting approaches have been proposed:

- The gradual deformation method allows to interpolate between spatial random fields to recast high-dimensional inverse problems into a sequence of one-dimensional problems (Hu, 2008).
- Sequential Gibbs sampling considers the sequential decomposition of the multivariate prior model density to control step sizes between prior samples (Hansen et al., 2012).

However, these approaches have been applied only to grid-based geological models with fixed geometry and resolution. A pioneering application of such concepts to geological structures was proposed by Cherpeau et al. (2012). More recently, level-set-based geometric parameterizations have received significant attention (Giraud et al., 2021), but have not yet been extended to consider faults and geological rules existing between geological models (Wellmann and Caumon, 2018). Finally, a recent class of approaches, termed Bayesian evidential learning (Scheidt et al., 2018), uses an ensemble of geological models and their input parameters and advanced statistical models to generate a functional relationship between the input models and the forecast. However, choosing the appropriate number of representative models to learn this relationship remains a challenge. Indeed, selecting too many models is computationally demanding because the forward problem must be solved on each model of the ensemble, while selecting too few models raises the risk of forecasting bias. To address this challenge, many characterization metrics have been proposed to cluster structural geological models based on geometric (Suzuki et al., 2008), topological (Lindsay et al., 2014; Pellerin et al., 2015; Pirot et al., 2022) or data-space criteria (Scheidt et al., 2015; Yin et al., 2020). All these distances, however, are computed after 3D geological models have been generated, which does not easily allow for exploring the search space.

In summary, the SaMoA project aims at discovering new ways to compute distances between geological structural models. The challenges stems from the unknown number of structures to use to explain a particular set of observations or make probabilistic forecasts, and from the complex interactions between geological structures (Wellmann and Caumon, 2018). SaMoA is expected to help addressing several overarching questions:

- Can we ensure that a sufficiently large set of prior geological structural models is considered to capture relations between model parameters and physical forecasts?
- Can we avoid wasting computational resources and minimize the redundancy between samples when exploring the prior geological parameter space?
- Can the sampling process use adaptive spatial resolution?

Approach

The SaMoA project (**S**ampling geological **M**odels **A**dsaptively for geophysical inverse problems) will primarily address sedimentary basins and focus on two classes of geological structures: stratigraphy and faults. Indeed, both types of objects span multiple spatial scales and induce significant anisotropy and complexity in the earth's properties. When considered in subsurface uncertainty studies, they are generally treated by generating realizations (obtained by independent random sampling from spatial data error distributions (e.g., Lindsay et al., 2014) or by geometric perturbations of a base interpretation (e.g., Suzuki et al., 2008). To reflect epistemic uncertainties, it is also possible to use a small set of expert-based scenarios (TODO: REFS). All these approaches may induce anchoring bias and understatement of uncertainties. In SaMoA, the modelling of both types of objects will be formalized by a combination of graphs and simple geological rules to quickly generate realizations of connectivity, and then geometric realizations using implicit surfaces. These two hierarchical levels of uncertainty can be integrated into a sequential Monte Carlo sampling.

The goal of the SaMoA project is to compute distances between stochastic geological models during the sampling of the faults and strata constituting these models. Such a computation will allow for:

1. Adapting the parameter space exploration strategy before and during the resolution of inverse problems (optimization by generating models which are close to existing ones, or exploration by enforcing repulsion between samples).
2. Considering spatial scale during the sampling process, so that the level of detail can be adaptively refined or coarsened during sampling, depending on the impact of a local change on the model forecast. This is essential both for matching observations and for quantifying prediction uncertainties.

Because sampling from non-informative prior space can make the resolution of inverse problems non practical, **the geological models produced by SaMoA will:**

- **Be anchored on spatial data deemed reliable** such as outcrop and borehole interpretations. The main idea will be to start from these data to define a relatively parsimonious search space exploring large-scale connection between the geological objects.
- **Honor geological correlation concepts formulated numerically**. For this, we will use graphs representing observations and geological correlation rules, based on our previous work on fault association (Godefroy et al., 2021) and well correlation (Caumon and Antoine, 2019).

Work Package: Integrating distances in the fault association sampler

Task 1.1: Design of a new sampling algorithm.

SaMoA will first revisit the fault association sampler (Godefroy et al., 2021) to integrate distances during simulation. In the original algorithm (Figure 3), the associations are processed sequentially, and each realization is independent from the others. Based on the association graph, distances between faults association scenarios will be defined, and the simulated associations will be organized in an efficient search data structure. This work will form the basis of a new Python library for the association of fault evidence. The critical parts of the code will be implemented in C++ for efficiency. A first version of the algorithm has already been prototyped (Oumar et al., 2022), and will need to be optimized for a large number of realizations. Several graph-based distances will also be investigated. This new sampler will

reuse the idea that the largest and most likely fault associations have a high probability to be sampled early in the simulation process (Godefroy et al., 2021).

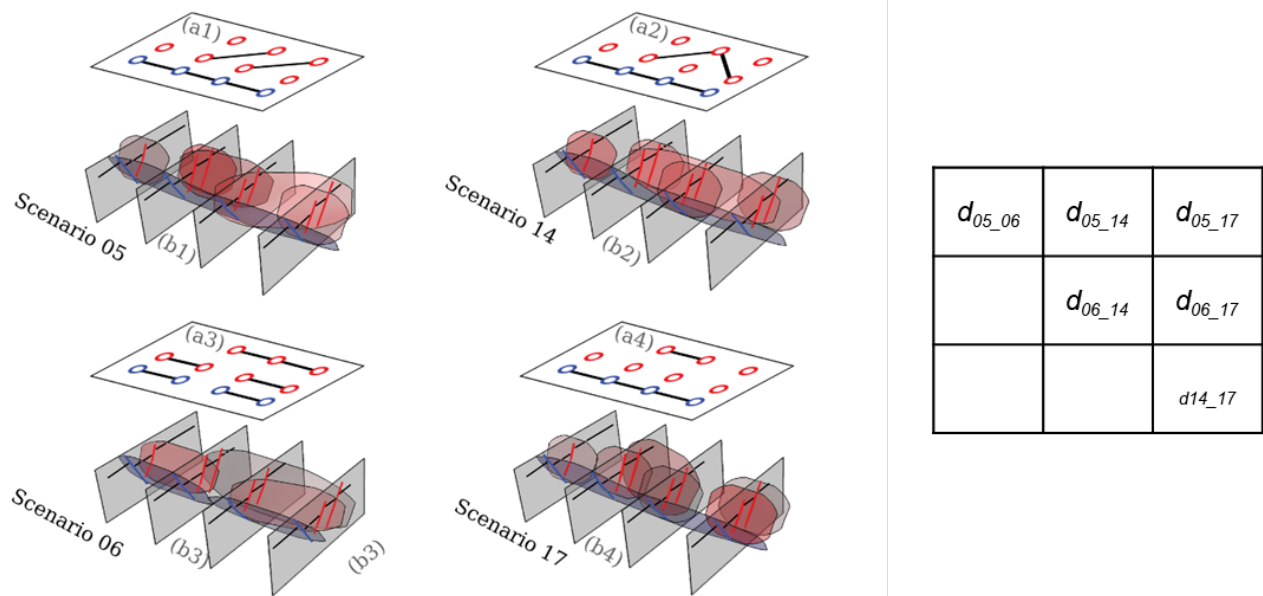


Figure 3 :Example of graph-based fault data associations (left, from Godefroy et al, 2021). The goal of SaMoA is to compute the distance matrix (right) between these associations during simulation.

Task 1.2: Assessment of the relations between fault association distances and model response.

In principle, large faults are expected to have the largest impact on the modelling outcome: large faults control the spatial organization of smaller faults, and are likely to define the large-scale compartments in aquifers or CO2 storage sites. This, however, will need to be checked and quantified for various problems. I plan to first study the relationship between the new topological association distances and static geomodel distances (Piriot et al., 2022), for which correlations should be significant. Then, I will consider the more challenging and risky question of how the association distances relate to the forecasting of subsurface flow. The choice of hydrodynamics problems is motivated by their relevance for the management of groundwater resources, geothermal energy and CO2 storage projects. As three-dimensional seismic data is seldom available for these projects, the proposed methodology aims to fill an important gap in uncertainty modelling. As faults can have arbitrary connectivity, the response for the candidate models will be computed on simulation codes able to deal with unstructured meshes, such as CSMP++ (Melbourne Uni / ETHZ), GEOSX (LLNL / Stanford / TotalEnergies) or Darts (Delft Uni). Because the insertion of finite faults in an unstructured mesh is not yet a solved problem, I have started discussing with the MMG inria team (Charles Dapogny, Algiane Frohely) and we plan to propose a joint PhD project on this project to the RING Consortium sponsors starting in 2023. Depending on the outcome of this potential PhD on fault insertion, an alternative will be consider other finite element meshers (e.g., SKUA-Gocad, tetgen, gmsh), or rasterization on fine-scale Cartesian grids. If correlations between the proposed model-space distance and the response can be found, we will also assess by how much they decrease when also sampling geometrical and petrophysical uncertainties for each association scenario. The absence of correlation may also be a possibility. In that case, the proposed methodology can still be used to address the ill-posedness of hydrodynamic inverse problems by proposing starting models for classical inverse methods based on a priori geological criteria only.

WorkPackage 2: Stochastic correlation of nested sedimentary sequences.

Task 2.1: Multi-scale stratigraphic correlation.

The second type of prior geological uncertainty addressed by SaMoA will concern multi-scale stratigraphic correlations, as the multi-scale aspect of sedimentary basins is widely recognized in sedimentology (Catuneanu, 2019; Pellegrini et al., 2020). In computer-assisted stratigraphic correlation, the notion of stratigraphic order has been considered as input for the correlation of two stratigraphic sections (Lallier et

al., 2016) or as a way to analyze multiple correlations a posteriori (Sylvester, 2022), but it has not yet been integrated satisfactorily into multi-well correlation. Therefore, a first goal of SaMoA will be to integrate stratigraphic scale in the multi-well correlation code WeCo (Caumon and Antoine, 2019). The WeCo code builds on the same principles as for multiple sequence alignment in bioinformatics (Higgins and Sharp, 1988) to merge correlated stratigraphic series incrementally and hierarchically using a 2D spatial search tree. The main idea is to extend the dynamic time warping method to higher dimensions without explicitly manipulating an intractable n-dimensional cost matrix, but the compromise is that the method may not necessarily find the optimal correlation. Therefore, WeCo propagates not the optimal correlation solution at each step, but the set of the N-best correlations. However, our findings so far suggest that this is not enough and that the obtained N-best solutions only explore the model space around a local minimum of the correlation cost function. Therefore, two aspects will be investigated in SaMoA:

- 1 The correlation order will be randomized to generate alternative correlation scenarios. Delaunay triangulations, edge flips and K-nearest-neighbor search strategies will be investigated for this randomization.
- 2 The stratigraphic scale will be integrated at each step using either deterministic input from sedimentologists or automatic order extraction using the continuous wavelet transform of the well logs (Hill et al., 2015). This will be performed by using the multi-well lower-order information results to constrain the higher-order correlation results.

This work will be done in collaboration with Christophe Antoine (research engineer at the RING Consortium). The obtained results will be validated on synthetic data sets to check the ability of the proposed algorithm to recover the reference correlation. The consistency of the stochastic set of correlations will be checked by considering how the ensemble of solutions evolves when new data are considered.

Task 2.2: Validation of the obtained correlations on a real data set.

A significant difficulty in any uncertainty model is the validation, as no such thing as a reference uncertainty exists. We have started applying multi-well correlations to true data in collaboration with Equinor, using biostratigraphic information as validation sets (Baville et al., 2022). In SaMoA, we will continue collaborating with Equinor on real-case applications of the developed approaches to assess in which case they produce results consistent with ancillary data. A possible target of interest could be the Sleipner CO₂ sequestration site or the new Norther light project, for which drilling data are already available¹, and where correlation with the nearby oil fields could help to better understand the interplay between tectonics and sedimentary processes and to assess the lateral facies variations and unconformities in the area. In complement to these subsurface applications, we also plan to start a new academic collaboration with Allard Martinus and Hemmo Abels (Delft U.). The target of this study will be lacustrine sediments of the Ebro Basin, where magnetostratigraphic data is available (Valero et al., 2014), and where the results of the stochastic correlation may allow to reinterpret the relative contributions of local tectonics versus climate forcings in the stratigraphic record. This study will be achieved in the frame of a two-year postdoc for which we will apply for competitive funding.

Task 2.3: Transdimensional inversion and stratigraphic correlation.

In the previous tasks, do consider the generation of stochastic correlations reflecting the subsurface uncertainties independently of their impact on the numerical forecasts of the subsurface's physical behavior. CF PhDs of Capucine and Julien.

Originality and risks of the project

Main benefits:

- Decoupling of connectivity and variable number of model parameters to represent prior structural knowledge
- Integrating spatial constraints in geophysical inverse methods (e.g., direct outcrop observations)
- Formulating geological priors in a mathematical / numerical way

¹ <https://www.equinor.com/news/archive/20201019-sharing-data-northern-lights>

- Generative approach
- Formulating sensible distance functions in model space (they may have only a poor correlation to the data space distances depending on the considered physics). The spatial sensitivity is also an important element: geological variations only matter in areas where the physics is sensitive. Even though this aspect should be considered for optimization purposes, it is not a priority for this project because the fundamental goal is precisely to assess geological uncertainties in areas where geophysics is blind, as this can help to define new data acquisition strategies.
- Prior geological models correspond to large sets and complex basis function in inverse problem terms, which may involve binary indicators (or Heaviside) functions and
- Multiple-point interactions and trade-offs between geological parameters (e.g., fault intersections, and stratigraphic correlations across faults).
- “Difficult to come up with a glorious uncertainty model”: need to communicate on cognitive biases, to and objective reasons to improve prediction uncertainty.
- How to manage / visualize / process a large set of realizations?
- Practical use in ensemble methods seem attainable, as incremental distance-based data analysis methods exist (Bengio et al., 2003).

Transfers

The SaMoA project will be realized in synergy with the RING Consortium, so the transfer aspects will be realized in this frame.

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Added value of the IUF and partnerships

The above project will be addressed in conjunction with the RING Consortium's partners. The RING Consortium is expected to provide a significant amount of funding for PhD students. We also anticipate that the interested RING partners will contribute with data to test the proposed methodologies.

The added value of the IUF is to alleviate my teaching load and give me more time to develop the fundamental and most challenging aspects of the problem. I also want and to make some numerical experiments to test some ideas.

9) POSSIBLE OPENING OF IUF PROJECT TOWARDS AN EUROPEAN PROJECT (ERC OR OTHER) DURING IUF DELEGATION, especially in the Proof of concept category

➔ **Maximum 2 pages** (cf. recommendations mentioned in the Application guidelines)

TODO

10) DESCRIPTION OF THEACHING ACTIVITIES, EDUCATIONAL AND ADMINISTRATIVE RESPONSABILITIES (MAXIMUM 2 PAGES)

Teaching activities during the last 5 years

In France:

- Home institution (number of teaching hours – calculated as equivalent to 1 hour of practical tutorials)
- In other institutions
- In other countries:

Educational and administrative responsibilities during the last 5 years

11) POSSIBLE OPENING OF IUF PROJECT TOWARDS A PROJECT OF INNOVATION IN TEACHING METHODS AND DISSEMINATION THROUGHOUT SOCIETY

➔ **maximum 1 page** (cf. recommendations mentioned in the Application guidelines)

12) LIST OF 3 FRENCH (2) OR FOREIGN (1) SCIENTIFIC EXPERTS WHO MIGHT BE CONTACTED BY THE JURY, in particular in the field of Innovation

☐ SIR

☐ MADAM

LAST NAME:

FIRST NAME:

POSITION:

UNIVERSITY AFFILIATION:

PHONE:

E-MAIL:

☐ SIR

☐ MADAM

LAST NAME:

FIRST NAME:

POSITION:

UNIVERSITY AFFILIATION:

PHONE:

E-MAIL:

☐ SIR

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